

Exploratory Data Analysis and Statistical Modeling of COVID-19 Epidemiology and Vaccine Portfolios across the EU/EEA (2020–2022)

Intelligent Data Analysis (DV1597) — Assignment 2

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Abstract. This paper presents an explanatory data analysis (EDA) and statistical modeling of the COVID-19 pandemic across 30 EU/EEA countries between 2020 and 2022. Using epidemiological databases from the European Centre for Disease Prevention and Control (ECDC), we investigate infection trajectories, vaccine brand distributions, and clinical capacity loads. We evaluate three main socio-epidemiological aspects: (i) the temporal relationship between cumulative vaccination rates and Case Fatality Rates (CFR) in Germany using time-lagged Pearson correlation; (ii) the transition in clinical severity (ICU-to-hospital occupancy ratios) from the pre-vaccine era (2020) to the post-vaccine era (2022) using a paired Wilcoxon Signed-Rank test; and (iii) the grouping of vaccine procurement strategies across countries using Principal Component Analysis (PCA) and K-Means clustering. Our statistical results show a significant reduction in clinical severity ($p = 0.0068$) and a strong negative correlation between second-dose coverage and rolling CFR ($r = -0.8236$), while highlighting key anomalies in international data reporting.

Keywords: COVID-19 · Epidemiology · Time-Lagged Correlation · Paired Inference · Unsupervised Clustering.

1 Introduction and Preprocessing Methodology

The ECDC-curated COVID-19 datasets contain daily reports of cases and deaths, weekly vaccination metrics by brand and demographic target groups, and daily/weekly hospital and ICU admission rates. However, direct aggregation of these datasets introduces severe statistical biases due to database structural anomalies and reporting discrepancies. To establish a sound analytical foundation, we implemented a unified data cleaning and population-level deduplication pipeline.

1.1 Standardization and Handling of Byte Order Marks (BOM)

The raw ECDC CSV files contain UTF-8 Byte Order Marks (BOM) in their headers (e.g., in the vaccination and cases datasets). If left unaddressed, standard column indexing fails. We ingested the datasets using the `utf-8-sig` encoding, standardizing all column headers and converting raw string dates into structured `datetime` formats covering the temporal window from January 1, 2020, to December 31, 2022.

1.2 NUTS-2 Regional Deduplication Strategy

A major structural challenge in the ECDC vaccination dataset is the co-existence of subnational sub-regions (categorized under the Nomenclature of Territorial Units for Statistics, NUTS-2) and country-level totals. Simple summation across all rows in the dataset results in massive double-counting. For example, summing all reported regional first doses for Austria yields a figure multiple times its actual population. To resolve this, we enforced a strict regional deduplication filter, retaining only records where the `Region` identifier is identical to the `ReportingCountry` ISO code (e.g., `Region == ReportingCountry`). This guarantees that only national-level metrics are aggregate-modeled.

1.3 Dynamic Under-18 Cohort Reconstruction

The vaccination dataset reports child demographics inconsistently across countries. While some nations (e.g., Italy, Germany) report vaccinations for the aggregate group `Age<18`, others (e.g., Sweden, Denmark, Netherlands) only report specific subsets such as `Age0_4`, `Age5_9`, `Age10_14`, and `Age15_17`. Simple exclusion of either category leads to skewed national rankings.

Table 1. Cumulative cases and population-level infection rates (2020–2022)

Country	Cumulative Cases	Population (2020)	Population Infected (%)
France	38,065,394	67,287,193	56.57
Germany	37,211,943	83,166,711	44.74
Italy	25,060,503	59,641,488	42.02
Spain	13,588,679	47,332,614	28.71
Poland	6,349,036	37,950,802	16.73
Netherlands	8,563,059	17,407,585	49.19
Romania	3,293,732	19,286,814	17.08
Belgium	4,691,499	11,521,238	40.72
Czechia	4,561,168	10,693,939	42.65
Portugal	5,539,949	10,295,909	53.81

To address this, we developed a dynamic cohort reconstruction algorithm. For each country, the pipeline evaluates the presence of **Age<18** in the reported target groups. If present, it uses this aggregate row as the sub-18 metric. If absent, it dynamically aggregates the specific child cohorts (**Age0_4** + **Age5_9** + **Age10_14** + **Age15_17**) to reconstruct the under-18 population coverage. This ensured a unified and unbiased demographic denominator.

2 Mandatory Questions and Empirical Findings (Section 3.1)

2.1 Q1: Top 10 Countries by Total Cases and Quarterly Wave Trajectories

Summing cumulative cases from 2020 to 2022, we identified the top 10 affected nations. To normalize these figures, we calculated the total case counts relative to the 2020 country population sizes to estimate the population-level infection rates (Table 1).

We subsequently tracked quarterly case totals to visualize epidemic wave patterns. The resulting trend line plot (**q1_quarterly_cases_trends.png**) reveals distinct epidemiological phases: a low-case suppression phase throughout 2020, moderate multi-wave surges during 2021 (Delta variant), and an explosive surge in Q1–Q2 of 2022 representing the transition to the highly transmissible Omicron variant, which dominated despite high vaccination coverage.

2.2 Q2: Spatio-Temporal Spatial Bubble Maps

To capture the geographic spread of the virus, we generated capital-centered spatial bubble maps for 2020, 2021, and 2022. By mapping cumulative cases and deaths, these figures visually demonstrate the transition of the pandemic’s gravity center. In 2020, Western Europe (France, Italy, Spain) bore the heaviest initial burden. By 2021, the burden expanded heavily into Central and Eastern Europe (Poland, Czechia, Romania). In 2022, the case bubble diameters expanded exponentially nationwide due to Omicron, while the corresponding death bubble diameters remained comparatively constrained, indicating a decoupling of transmission from mortality.

2.3 Q3: Vaccine Brand Popularity and Country-Level Exceptions

Analyzing national vaccination data (filtering out subnational double-counting), the total doses administered (combining primary and booster doses) across the EU/EEA revealed a highly consolidated market. The three most dominant vaccine brands overall are:

1. **Comirnaty (Pfizer/BioNTech) [COM]:** 60.3% overall share.
2. **Spikevax (Moderna) [MOD]:** 13.4% overall share.
3. **Vaxzevria (AstraZeneca) [AZ]:** 12.2% overall share.

However, systematic brand-exception mapping identified key national anomalies:

- **Liechtenstein [LI]:** Deviated dramatically from the Pfizer-dominated regional norm, administering Spikevax (Moderna) as its primary brand (65.5% share) and Comirnaty (31.9% share).
- **Germany [DE]:** In the ECDC database, brand-specific reporting for Germany is absent; all doses are categorized under the code **UNK** (Unknown Vaccine), representing 100.0% of its reported vaccination entries. This presents a critical reporting outlier.
- **Hungary [HU]:** Leveraged alternative portfolios including BBIBP-CorV (Sinopharm) and Sputnik V, bypassing standard EU centralized procurement.

2.4 Q4: Target Demographic Groups for Top 3 Vaccine Brands

Breaking down doses by target groups (excluding aggregate categories such as ALL and 1_Age60+), we mapped the target demographics of the top 3 vaccines. Comirnaty (Pfizer) exhibited a universal profile, distributed heavily across all age brackets from children (Age<18) to elderly (Age80+) and healthcare workers (HCW). Spikevax (Moderna) was oriented predominantly toward older adult cohorts (Age50_59, Age60_69, Age70_79). In contrast, Vaxzevria (AstraZeneca) was target-focused toward middle-aged cohorts (Age25_49, Age50_59) and healthcare professionals, reflecting changing clinical guidance and age-specific safety recommendations during the 2021 rollout.

2.5 Q5: Socio-Epidemiological Analysis - Vaccine Skepticism and Peak Hospitalization

To evaluate the impact of vaccine skepticism on clinical healthcare systems, we defined a standardized national ****Vaccine Skepticism Rate**** ($1 - \text{First Dose Coverage}$) and merged it with the country-level peak weekly new hospital admissions per 100,000 population during 2021.

We performed a Pearson Correlation test on the merged dataset. The analysis yielded a Pearson correlation coefficient of:

$$r = 0.5414 \quad (p = 0.0166)$$

This represents a statistically significant, moderate-to-strong positive correlation. Countries with higher vaccine skepticism (e.g., Bulgaria, Romania) experienced significantly higher peak weekly hospitalization burdens during 2021 compared to highly vaccinated countries (e.g., Ireland, Portugal), validating the real-world protective efficacy of vaccine coverage in reducing clinical load.

2.6 Q6: Under-18 First-Dose Vaccination Ranks

Using our dynamic child cohort reconstruction, we ranked countries by first-dose vaccinations administered to the under-18 population relative to the country's total population. The analysis showed wide discrepancies:

- **Top 3 Countries:** Italy (7.3%), France (6.8%), and Luxembourg (6.8%).
- **Bottom 3 Countries:** Bulgaria (0.4%), Croatia (0.9%), and Romania (1.4%).

This highlights a distinct East-West divide in pediatric and adolescent vaccination policies and public acceptance.

2.7 Q7: Weighted Average Age of Second-Dose Recipients

Calculating the weighted average age of second-dose recipients across the 30 countries required a robust statistical fallback for Germany. Germany only reports coarse target groups (Age<18, 1_Age<60, and 1_Age60+) rather than granular cohorts. If processed through the standard pipeline, all adult rows are filtered out as aggregate groups, skewing Germany's average age to a misleading 9.0 years (representing only children).

Our algorithm resolved this by checking if a country reports granular adult cohorts. If not, it falls back to a coarse midpoints map: Age<18 (9.0 years), 1_Age<60 (39.0 years), and 1_Age60+ (72.5 years).

Under this unified model, the oldest second-dose populations were in Bulgaria (53.7 years), Croatia (52.7 years), and Slovenia (52.3 years). The youngest were in Ireland (42.2 years) and Spain (44.0 years). Germany was calculated at a realistic 48.6 years, demonstrating the necessity of the fallback logic.

2.8 Q8: Clinical Capacity and Healthcare Burden Transitions (2020 vs. 2022)

We compared the peak daily ICU and hospital occupancy per 100,000 population between the pre-vaccine 2020 waves and the post-vaccine 2022 Omicron waves. Across almost all major nations (e.g., Belgium, Czechia, France, Netherlands), peak daily ICU occupancy per 100k dropped significantly. For instance, Belgium's peak daily ICU occupancy fell from 12.8 per 100k (2020) to 4.5 per 100k (2022). This clinical decoupling occurred despite 2022 having exponentially higher case numbers, providing powerful visual evidence of vaccine-mediated protection against critical illness.

3 Advanced Statistical Modeling and Inference (Section 3.2)

3.1 Advanced Q1: Socio-Epidemiological Coupling - Dynamic Lagged CFR Correlation

To explore how the vaccine rollout dynamically coupled with clinical mortality in Germany, we calculated the rolling weekly Case Fatality Rate (CFR) using a 4-week window to account for temporal fluctuations:

$$\text{CFR}_t = \frac{\sum_{i=0}^3 \text{Deaths}_{t-i}}{\sum_{i=0}^3 \text{Cases}_{t-i}} \times 100$$

We then performed time-lagged Pearson correlation tests against the cumulative second-dose vaccination rate, shifting the CFR series forward by $L \in [0, 8]$ weeks to model the therapeutic delay between vaccine administration and mortality suppression.

The empirical lagged correlation results are presented in Table 2.

Table 2. Time-Lagged Pearson Correlation: Vaccine Coverage vs. CFR (Germany)

Lag (L , Weeks)	Pearson Correlation (r)	p-value
0 (Co-temporal)	-0.8236	< 0.0001
1 Week	-0.8056	< 0.0001
2 Weeks	-0.7891	< 0.0001
3 Weeks	-0.7719	< 0.0001
4 Weeks	-0.7560	< 0.0001
6 Weeks	-0.7323	< 0.0001
8 Weeks	-0.7311	< 0.0001

The analysis reveals an extremely powerful, highly significant negative correlation ($r = -0.8236$ at lag 0, maintaining a strong $r = -0.7311$ at lag 8). This strong inverse coupling quantifies the massive, sustained protective effect of the vaccination campaign, showing that as second-dose coverage expanded, the baseline lethality of the virus collapsed systematically in Germany.

3.2 Advanced Q2: Paired Clinical Severity Transitions - Wilcoxon Signed-Rank Inference

We tested the hypothesis that clinical severity shifted between 2020 and 2022. We defined the clinical severity ratio as the average daily ICU occupancy divided by the average daily hospital occupancy per country:

$$\text{Severity Ratio}_c = \frac{1}{T} \sum_{t=1}^T \frac{\text{ICU Occupancy}_{c,t}}{\text{Hospital Occupancy}_{c,t}}$$

Since severity ratios are bounded, non-negative, and non-normally distributed, standard parametric paired t -tests are inappropriate. We applied the non-parametric **Wilcoxon Signed-Rank Test** to compare the paired country ratios between 2020 and 2022.

The test yielded:

$$W = 4.0 \quad (p = 0.0068)$$

Since $p < 0.01$, we reject the null hypothesis. The clinical severity ratio decreased highly significantly across the EU/EEA (e.g., Austria’s average severity ratio dropped from 23.0% to 9.2%; the Netherlands’ fell from 33.9% to 10.2%). This statistically confirms that patients hospitalized in 2022 required intensive care at a fraction of the rate observed in 2020, representing a massive shift in clinical severity.

3.3 Advanced Q3: Geopolitical Brand Portfolios - PCA and K-Means Clustering

To explore international vaccine procurement and brand structures, we normalized country-level vaccine brand distributions into share-proportions and applied **Principal Component Analysis (PCA)** for dimensionality reduction, coupled with **K-Means Clustering** ($K = 3$) for unsupervised grouping.

The PCA loading vectors reveal that:

- **PC1 (Explains 64.9% Variance):** Driven negatively by Comirnaty (Pfizer) [COM] loading (-0.597) and positively by Unknown Vaccine [UNK] loading (0.795).

- **PC2 (Explains 23.1% Variance):** Driven heavily by Spikevax (Moderna) [MOD] loading (0.860) and negatively by Comirnaty (−0.403).

This projection partitioned the 30 countries into three distinct geopolitical clusters (visualized in `adv_q_vaccine_pca_clusters.png`):

1. **Cluster 1 (Standard European Portfolio - 28 countries):** Includes AT, BE, BG, ES, FR, IT, SE, etc. These countries relied heavily on centralized EU procurement, exhibiting a highly integrated brand composition: Comirnaty (Pfizer/BioNTech) dominant (mean 60.3%), Spikevax (Moderna) secondary (mean 13.4%), and Vaxzevria (AstraZeneca) tertiary (mean 12.2%).
2. **Cluster 2 (Moderna Outlier - 1 country):** Liechtenstein [LI]. Positioned in the upper quadrant of PC2 due to its unique reliance on Spikevax (Moderna) as its primary brand (65.5% share).
3. **Cluster 3 (Unknown/Reporting Outlier - 1 country):** Germany [DE]. Placed as an extreme outlier in PC1 due to its unique ECDC database formatting, reporting all doses under the code UNK (100.0% share).

4 Discussion, Database Limitations, and Conclusion

4.1 Epidemiological and Geopolitical Insights

Our statistical models reveal that the COVID-19 pandemic’s clinical burden was highly malleable. The strong inverse coupling between Germany’s vaccination rate and CFR ($r = -0.8236$) and the significant paired decline in ICU-to-hospital occupancy ratios ($p < 0.01$) demonstrate that vaccine rollout and Omicron’s emergence successfully decoupled case transmission from mortality and intensive care capacity. Geopolitically, the PCA clustering reveals that the EU centralized vaccine procurement succeeded in delivering a highly uniform portfolio across 28 of the 30 nations, while exposing unique localized procurement (Liechtenstein) and reporting (Germany) variations.

4.2 Critical Database Schema and Reporting Limitations

Biostatistical modeling on the ECDC datasets revealed several critical data-quality limitations:

- **Database Formatting Discrepancies:** Germany’s categorisation of all doses under UNK and its failure to report granular age cohorts represents a major data-sharing gap that severely restricts comparative national research.
- **Subnational Double-Counting Risks:** The co-existence of NUTS-2 regional rows and national totals in the vaccination dataset creates a high probability of analytical error. Public health dashboards must separate regional and national tables to prevent severe double-counting.
- **Testing Policy Shifts:** Cumulative case numbers in 2022 are heavily biased by shifts in testing policies (transitioning from universal PCR testing to rapid antigen home testing and restricted PCR eligibility), artificially lowering case counts relative to 2020.

4.3 Conclusion

This study highlights the power of intelligent data analytics in extracting actionable epidemiological insights from complex, noisy datasets. By implementing robust data cleaning, regional deduplication, and dynamic cohort fallback algorithms, we successfully modeled vaccination, capacity, and severity trends across the EU/EEA. Our findings provide mathematical validation of the protective power of vaccine campaigns and variant transitions, while detailing the vital importance of data standardization in global health informatics.